Monitoring Tools

Why Monitoring Tools

- To continuously monitor servers and applications
- Automated alerts generation when the event volates the threshold
- Enables 24*7 monitoring team to watch the infrastructure and application performance continuously

Infrastructure Monitoring

Zenoss
Solar winds
HP Openview
IBM Tivoli
CA Unicenter
Nagios

Application Monitoring

Sitescope
Business Service Management (BSM)
Dynatrace
Appdynamics
New Relic APM

Note: Tools like Splunk, ELK and Kibana can monitor the system and application log files as well.

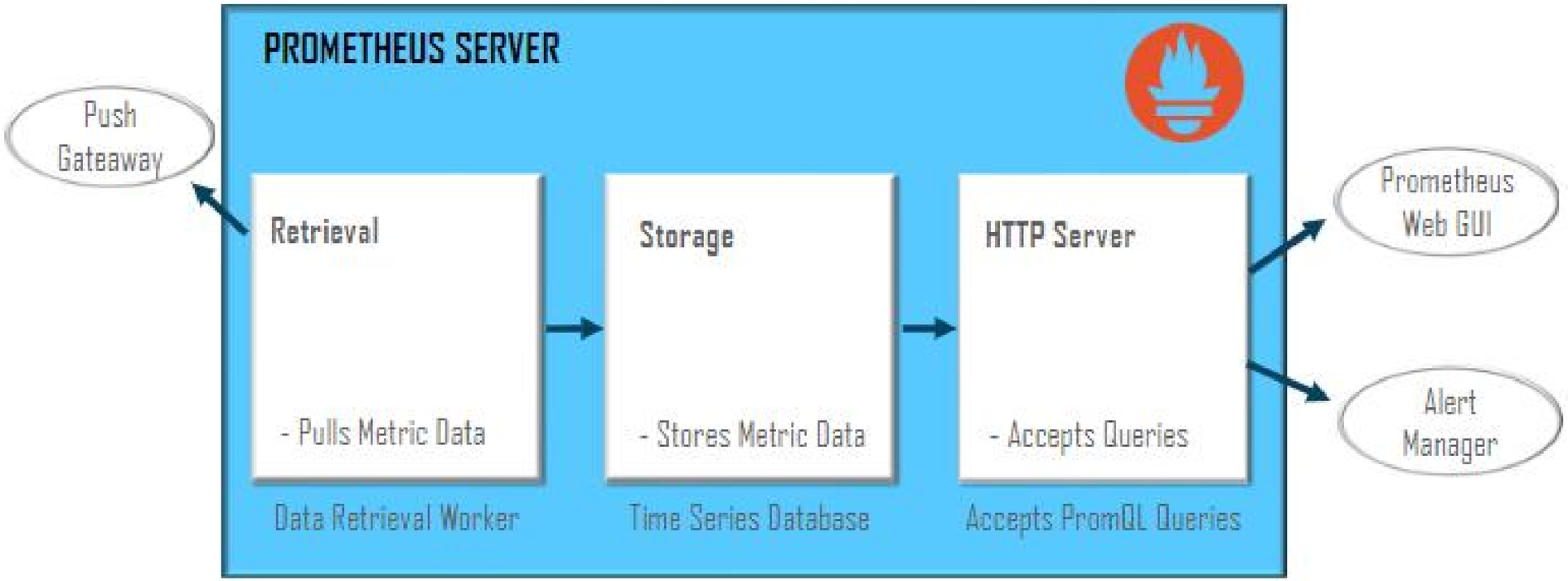
How Monitoring Tools work

- Can be agentless or agent-based
- Agents collect metrics from respective and posts the information to the manager. Manager sends alert notifications to the monitoring team.
- Monitoring tools can integrate with Ticketing tools like Service Now and open a auto ticket in Service Now whenever an event violates the thresold.

Prometheus

- Open source systems monitoring and alerting toolkit
- Collects and stores metrics as a time series data
- Scrapes targets
- PromQL is the language used to query time series in Prometheus
- Service discovery helps find our services and monitor them
- Exporters help to monitor 3rd party components
- Can send alerts to alert manager
- Prometheus runs on port 9090 and alert manager on 9093

Prometheus Architecture



What does Prometheus monitor?

- Linux/Windows Server
- Apache Server
- Single Application
- Service, like Database

Which units are monitored of those targets?

- CPU Status
- Memory/Disk Space Usage
- Requests Count
- Exceptions Count
- Request Duration

Monitoring your applications

- Load on the application (total number of requests)
- Response time
- Errors and Exceptions

Using client libraries you can expose /metrics endpoint

Grafana

- Multi-platform open source analytics and interactive visualization web application
- Provides charts, graphs and alerts for the web when connected to the supported data services
- Some data sources supported by Grafana: AWS Cloud watch, Azure monitor, Postgre SQL, Elastisearch etc..
- Grafana dashboards can be personalized as per our requirements.

Prometheus Links

https://prometheus.io/docs/introduction/overview/

https://prometheus.io/docs/concepts/metric_types/

https://chronosphere.io/learn/an-introduction-to-the-four-primary-types-of-prometheus-metrics/

https://prometheus.io/docs/instrumenting/exporters/

https://prometheus.io/docs/instrumenting/clientlibs/